Vision and Scope Document
for
docWise

Version 1.1 approved

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Docwise Organization
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Revision History

Below is the **update history of project documents**, including the reasons for changes and version information.

Name	Date	Reason for Changes	Version
Rukiye Sedef Keskin	01/03/2025	The first version was created, and the project documents were completed.	1.0
Rukiye Sedef Keskin	04/05/2025	The scope of the project has changed, and changes are reflected on functionalities.	1.1

1. Business Requirements

This section defines the business requirements of the AI-assisted diagnosis system and the reasons for developing the project. The business requirements cover the values the system will provide, business opportunities, success criteria, and potential risks.

1.1. Background

Medical misdiagnoses caused by human error in diagnostic processes can lead to serious health issues. Globally, **approximately 20% of patients are misdiagnosed**, and **95% of these misdiagnoses are due to human factors**. In particular, **women and ethnic minorities are disproportionately affected**, making up **30% of misdiagnosed patients**. Furthermore, in some malicious practices, unnecessary treatments are applied to patients, putting their health at risk.

To address these issues, **docWise** has been developed as an AI-powered medical diagnosis and decision support system. It will operate as a **model committed to ethical values, free from discrimination, and trained on extensive datasets**. This aims to make the diagnosis and treatment processes among patients, doctors, hospital administrators, and health authorities more reliable, transparent, and accountable.

The system includes secure user authentication via e-Devlet, integration with the e-Nabız platform for automatic transfer of health data, AI-based diagnostic suggestions for doctors, graphical health analyses and lifestyle recommendations for patients, compliance analysis for administrators, and audit functions for the Ministry of Health Inspection Board. The diagnostic suggestions generated by the AI can be approved or rejected by doctors with justification.

1.2. Business Opportunity

In today's healthcare sector, making accurate diagnoses is a critical factor that directly affects patients' treatment processes. However, due to time constraints, lack of experience, or conscious/unconscious errors by physicians, serious mistakes can occur during the diagnostic process.

This project aims to make a significant contribution to the healthcare industry by developing an AI-powered diagnostic assistant that ensures patients receive accurate and timely diagnoses. By utilizing a model trained on large-scale datasets, the system will support doctors in the decision-making process and reduce diagnostic errors—particularly those affecting women and ethnic minorities.

Furthermore, **docWise** not only strengthens the doctor-patient relationship but also provides essential decision support functionalities for other healthcare stakeholders such as hospital administrations and the Ministry of Health's Inspection Board. With its versatile structure, the system aims to improve the quality of healthcare services while offering faster, more reliable, and cost-effective solutions—creating a competitive advantage in the sector.

This system can be adopted across various domains, including hospitals and telemedicine platforms, offering a superior alternative to traditional manual diagnostic methods through its speed, reliability, and accuracy.

1.3. Business Objectives and Success Criteria

The main business objectives of this project are:

- **Reducing misdiagnosis rates by more than 50%**: The AI-assisted diagnosis system will minimize error rates by being trained on extensive datasets.
- **Eliminating gender- and ethnicity-based diagnostic discrimination**: The system will operate with neutral and objective data to prevent bias.
- **Providing fast and accurate diagnoses**: It will produce faster and more effective results compared to traditional diagnostic methods.
- **Enhancing patient safety**: It will prevent unnecessary treatments and protect patient health.
- **Economic sustainability**: By reducing diagnostic costs in the healthcare sector, the system will become more accessible to a greater number of patients.
- **Ensuring holistic data use in healthcare services:** Supporting effective use of patient history and medical data by working in integration with national systems such as E-Nabız.

The following measurable criteria will be used to evaluate the success of these objectives:

- The accuracy rate of AI diagnoses should be **75% or higher**.
- There should be **a 40% reduction** in complications caused by misdiagnosis.
- The system should be accepted as a **pilot application by at least 50 hospitals or clinics**.

1.4. Customer or Market Needs

The increasing number of patients in the healthcare sector increases the burden on doctors. In current systems, doctors have limited time and information when making diagnoses. In addition, some patients are subject to discrimination during the diagnostic process.

This project was **developed to accelerate diagnostic processes and enable doctors to make more informed decisions**. Since the AI-supported system is trained **with larger data sets, it will minimize human error and allow doctors to work more efficiently.**

Existing solutions in the market mostly work with limited medical data. However, this project will differentiate itself from its competitors by increasing diagnostic accuracy with advanced machine learning algorithms. In addition, patients will be able to better monitor their own health status with a mobile or web-based version to be developed for individual users**. The patient will be**

able to receive graphical analyzes of their health status and receive life support recommendations.

1.5. Business Risks

The potential business risks encountered during the development of this project are:

- **Data Access and Privacy**: The AI model will require comprehensive patient data for training. However, legal regulations regarding data security and patient privacy must be considered.
- **Doctor Resistance**: The integration of AI into diagnostic processes may face resistance from some doctors. It should be explained that AI serves as a supportive tool, and efforts should be made to encourage doctors to adopt this technology.
- **Regulations and Ethical Guidelines**: The use of AI in the healthcare sector is subject to strict regulations in many countries. The developed system must comply with legal and ethical standards.
- **Risk of Misdiagnosis**: The AI model's decisions may not always be accurate. Therefore, continuous updates to the model and collaboration with doctors are necessary.
- **Market Acceptance**: The adoption of new technologies in the healthcare sector can take time. For this reason, initial pilot tests should be conducted in specific clinics and hospitals to demonstrate the system's reliability.

The following measures will be taken to minimize these risks:

- **Anonymization of patient data** and compliance with laws such as KVKK and HIPAA.
- **Training programs for doctors**, explaining that AI systems serve as supportive tools to reduce workload.
- **By providing access to diagnosis and treatment plans,** patients should be supported to make more informed decisions in their healthcare processes.
- **Development in accordance with legal and ethical guidelines.**

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2. Vision

This section defines the long-term vision of the AI-assisted diagnosis system. The vision explains how the developed system will transform in the future to make diagnostic processes in the healthcare sector safer, more transparent, and ethical.

2.1. Vision Statement

The AI-supported diagnostic system, which will be developed to ensure that patients receive **accurate, fair and reliable diagnoses**, will be a health innovation that **minimizes human error**, eliminates discrimination and provides a transparent diagnostic process.

This system will provide doctors with an intelligent assistant in the diagnostic process, and will receive life support recommendations by viewing patients' diagnoses and treatment plans. In the event that the doctor rejects the diagnosis, it will be mandatory to provide a clear justification to the system.

This system will provide data-driven health analyses not only for individual patients but also for hospitals and public health institutions, allowing for better diagnostic and treatment planning.

2.2. Major Features

User Registration and Authentication Processes

1. The patient registers in the system via the "Patient Registration" option using the e-Devlet platform.
2. The patient logs into the system via the "Patient Login" option using the e-Devlet platform.
3. The doctor registers in the system via the "Doctor Registration" option using the e-Devlet platform.
4. The doctor logs into the system via the "Doctor Login" option using the e-Devlet platform.
5. The administrator logs into the system via the "Admin Login" option.

User Interface Functions

6. Users can update their registered phone number and email address in the system.
7. Users can select the user interface theme (e.g., dark mode).
8. Users can choose the interface language preference (Turkish-English).

Patient Functions

9. Patients link their e-Nabız account to automatically transfer health data into the system.
10. Patients view their previous diagnoses and treatment plans.
11. Patients download their previous diagnoses and treatment plans as PDF files.
12. Patients receive graphical analyses based on their historical health data.
13. Patients receive AI-based life support suggestions based on graphical analyses.
14. Patients upload their current insurance policy to the system.

Hospital Management Functions

15. For public hospitals, hospital management registers the patient's appointment.

16. For private hospitals, hospital management generates pricing based on the patient's insurance policy coverage for the appointment.
17. For private hospitals, hospital management registers the patient's appointment upon acceptance of the pricing.
18. For public hospitals, hospital management creates a laboratory test record for the patient.
19. For private hospitals, hospital management enters the requested tests and generates pricing based on the patient's insurance policy coverage for the tests.
20. For private hospitals, hospital management creates a laboratory test record upon acceptance of the pricing.
21. For public hospitals, hospital management transmits all uploaded test data to the e-Nabız system.

Doctor Functions

22. Doctors view the list of patients.
23. Doctors select the next patient from the list and initiate the treatment process.
24. Doctors enter the patient's current symptoms into the system.
25. If necessary, doctors request laboratory tests for the patient through hospital management.
26. Doctors view the test results uploaded by the hospital laboratory.
27. Doctors select relevant parameters (symptoms, test results, imaging results, etc.) to request a diagnosis from the AI module.
28. Doctors view the AI's diagnostic suggestion.
29. Doctors approve the AI's diagnostic suggestion.
30. Doctors reject the AI's diagnostic suggestion with a justified reason.
31. Doctors prepare the patient's treatment plan and submit it to the system.

Artificial Intelligence Model Functions

32. The AI model provides life support suggestions based on the patient's graphical analysis.
33. The AI model analyzes the parameters submitted by the doctor and provides a diagnosis.

Admin Functions

34. Admin view the list of registered doctors and select a doctor.
35. Admin view the selected doctor's examination list for the past three months and select an examination.
36. Admin view the details of the selected examination (patient tests, AI and doctor's diagnoses).
37. Admin view the treatment plan determined by the doctor for the patient.
38. Admin analyze the necessity of the patient's treatment plan and enter the analysis into the system.
39. The AI model generates and displays statistics on diagnostic accuracy based on the doctor's negative feedback and findings.

40. Admin retrain the AI model with new data using the doctor's feedback to improve model accuracy.
41. Admin measure the compatibility between doctors' diagnoses and the AI's suggestions.
42. Admin identify treatment plans with low necessity and add the doctor to the investigation list.
43. Admin view the current investigation list and select a doctor.
44. Admin review the selected doctor's previous tests, diagnoses, and treatment plans.
45. If more than 30% of the reviewed doctor's treatment plans are deemed unnecessary, admin flags the doctor with a warning as malicious.
46. Admin forward the doctor's diagnosis and treatment plan, along with test results, to the Ministry of Health's Inspection Board.

Hospital Laboratory Functions

47. Laboratories view the list of test requests received by the system.
48. Laboratories select the next patient from the list and initiate the testing process.
49. Laboratories upload test results to the system.

Ministry of Health Inspection Board Functions

50. The inspection board views the list of doctors flagged with warnings.
51. The inspection board reviews the details of cases involving flagged doctors.
52. The inspection board suspends system access for doctors decided to be prosecuted.

2.3. Assumptions and Dependencies

For the successful implementation of this project, certain assumptions and dependencies must be met:

- **Accurate and Comprehensive Datasets**: The AI model will be trained using extensive and accurate diagnostic data. Collaboration with healthcare institutions and service providers will be required.
- **Regulatory and Ethical Compliance**: The system will be developed in full compliance with KVKK, HIPAA, and other health data privacy regulations.
- **Doctors' Adaptation to AI-Assisted Work**: Training and awareness programs must be conducted to ensure that doctors see AI as a supportive tool.
- **Integration with Healthcare Institutions and Hospitals**: API and data integration must be ensured for the AI system to operate seamlessly with healthcare institutions.
- **Continuous Updating of AI Algorithms**: The algorithms must be continuously improved with new data and feedback to enhance diagnostic accuracy.
- **Increasing Patient Adoption Rates**: To gain patient trust in the AI-assisted diagnosis system, a clear, understandable, and secure user experience must be provided.
- **Technological Infrastructure Requirements**: Running AI models that require high processing power **necessitates integration with cloud-based or local data centers.**

Following these assumptions, technological, ethical, and operational processes will be designed to provide a sustainable and successful solution.

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3. Scope and Limitations

This section defines the **scope and limitations** of the AI-assisted diagnosis system. It clarifies which features the proposed solution will include and which elements will remain outside the system. This helps stakeholders set realistic expectations. Additionally, it provides a reference framework for evaluating requirement changes.

3.1. Scope of Initial Release

The first version of the product will include **core features that support diagnostic processes in the healthcare sector** and make the diagnosis process between patients and doctors more transparent. The initial version will include the following key components:

- **AI-Assisted Diagnosis System**: An AI model that provides possible diagnoses based on patients' symptoms and test results (X-ray, blood tests, urine tests, MRI, etc.).
- **Doctor Support Module**: A system that assists doctors by providing AI-generated diagnostic recommendations to support decision-making.
- **Diagnosis Rejection Justification System**: A mechanism ensuring that doctors provide justifications when rejecting an AI diagnosis.
- **Patient Medical History Integration**: Data analysis that enhances diagnostic accuracy by incorporating patients' health history.
- **Real-Time Data Analysis**: Continuous AI learning and updates with new medical data.
- **Web-Based Patient Access**: Allowing patients to view their diagnosis information through a web interface.

These features are designed to provide the **greatest value to the widest range of patients and doctors**. In the first version, **core diagnostic capabilities and user experience optimization will be prioritized**.

3.2. Scope of Subsequent Releases

The following are **advanced features and integrations** planned for future versions, which are not included in the initial release:

- **Advanced Personalization**: Intelligent analytics that provide recommendations based on users' medical history and lifestyle.
- **Doctor-Doctor Collaboration Module**: Sharing mechanisms that enable different doctors to collaborate on AI-assisted diagnoses.
- **Telemedicine Integration**: A digital health infrastructure that enables patients to receive remote diagnosis and consultation services.

These additions will be **gradually developed to expand the system's reach and better respond to user needs**.

3.3. Limitations and Exclusions

Some stakeholders may expect the following features; however, these are **not planned** for the initial release or future versions:

- **Automated Treatment Decisions**: AI will only assist in the diagnosis process and **will not** directly recommend treatment plans.
- **Self-Diagnosis and Treatment**: The system is designed to be **a support tool exclusively for doctors and healthcare professionals**. It is **not** intended for patients to diagnose themselves or prescribe medications.
- **A Universal Model Covering All Diseases**: Initially, the system will focus on specific disease groups and will expand gradually over time.
- **Integration with Medical Devices in Hospitals**: In the first version, there will be **no direct integration** with medical devices used in hospitals.
- **Data Sharing Among Patients**: Patient data will be **kept completely confidential**, and there will be **no data-sharing feature** between patients.

These limitations have been set to ensure **the secure and ethical** development of the system. Throughout the development process, these restrictions will be reviewed based on **user feedback and technical feasibility**.

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4. Business Context

This section defines the **business context** of the AI-assisted diagnosis system. It focuses on identifying **the target customer groups**, determining which **business processes** will be affected, and how **management priorities** will be established.

4.1. Stakeholder Profiles

Stakeholder	Major Value	Attitudes	Major Interests	Constraints
Patients	Diagnosis and plan follow-up	Expect a user-friendly system, prioritize data privacy	AI supported healthy life support recommendation system	Data privacy concerns
Doctors	Reduction in misdiagnosis rates	Open to AI assistance but wants to make the final decision	Faster and more accurate diagnosis	Need for integration with existing systems

Stakeholder	Major Value	Attitudes	Major Interests	Constraints
Hospitals & Clinics	Reduction in operational costs	Open to efficiency improvements but hesitant about costs	Integration with hospital record systems	Budget constraints, need for staff training
Government & Regulatory Bodies	Improvement of public health policies	Supportive but require full adherence to ethical regulations	Data collection for accuracy analysis of AI-assisted diagnoses	Compliance with data protection and health laws
Developers	Sustainability of technical infrastructure	Require well-documented APIs and clear documentation	Ease of integration, scalability, performance optimization	**Minimum budget of $3M**

4.2. Project Priorities

Dimension	Target Objective	Constraint	Degree of Freedom
Schedule (Time Plan)	The system must start development on March 1, 2025, and be completed by June 1, 2025.	Delay period can be a maximum of **2 weeks**.	Additional resources may be allocated to accelerate the process.
Features (Functionalities)	Critical features such as AI-assisted diagnosis, doctor/patient interface, data security, and model validation must be completed.	At least **60%** of critical features must be included in **v1.0** release.	Additional features may be added after the priority features are completed.
Quality	The system must be highly reliable for doctors and patients.	User acceptance tests must achieve a **minimum success rate of 75%**.	If test success reaches **75%+**, additional development time may be granted.
Staff	The entire development process will be carried out by a dedicated team.	**Time management** is crucial throughout the project.	External support (consulting, API services, etc.) may be obtained if needed.
Cost	The project cost must be kept at a minimum level.	Additional hardware or external services must be limited to a **5%** budget increase.	A budget deviation of **up to 15%** can be accepted without requiring further

Dimension	Target Objective	Constraint	Degree of Freedom
			approval.

4.3. Operating Environment

The system is intended for global use by patients, doctors, hospitals, and regulatory authorities. Therefore, it must meet high standards of **accessibility**, **reliability**, **performance**, and **data integrity**.

- **Geographical Distribution and Time Zones**
  - Users will access the system from different geographical regions.
  - The infrastructure must support multiple time zones to ensure global usability.

- **Access Scheduling**
  - Patients should be able to access the AI-assisted system and their medical history at any time.
  - Doctors must be able to connect to the system in real time during patient appointments and receive diagnostic support.
  - Hospitals and healthcare administrators should have continuous access to system analytics and diagnostic trends.

- **Data Management and Storage**
  - Patient data will be stored on both local and cloud-based servers.
  - Data integrity will be maintained, and information from different locations will be merged into a centralized structure.
  - Critical data, such as patients' medical history, diagnostic records, and doctors' feedback, will be updated using real-time data synchronization.

- **System Performance and Response Times**
  - The AI diagnosis system must analyze real-time data and provide a diagnostic recommendation **within a maximum of 10 seconds.**
  - The latency for accessing remotely stored data should be no more than **3 seconds.**
  - The system must be optimized with high-performance servers and distributed databases.

- **System Availability and Fault Tolerance**
  - Uninterrupted system operation is critical for patients and doctors.
  - For emergency services and intensive care units, the system must guarantee **99.9% uptime.**

- Backup and fault tolerance mechanisms must be in place to prevent system crashes.

- **Access Security and Data Protection**
  - Full compliance with **GDPR, HIPAA, and KVKK** health data privacy laws must be ensured.
  - **E-devlet system integration** will be mandatory for **patient and doctor logins**.
  - **Patient data will be stored using cryptographic encryption** and will only be accessible to authorized personnel.
  - Access logs will be monitored regularly to detect unauthorized access attempts.

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5. Risk Impact, Probability and Owners

Risk Level	Probability (%)
High (H)	70%+
Medium (M)	30% - 70%
Low (L)	Less than 30%

- **High Impact (H)** → Significantly disrupts the core functionality of the system.
- **Medium Impact (M)** → Causes disruptions in certain system functionalities.
- **Low Impact (L)** → Creates minor issues but does not affect core system performance.

- **Owners**
  - **Product Manager (PM):** Rukiye Sedef Keskin
  - **Configuration Manager (CM):** Rukiye Sedef Keskin
  - **Software Architect (SAr):** Rukiye Sedef Keskin
  - **Software Analyst (SAn):** Rukiye Sedef Keskin
  - **Software Tester (ST):** Rukiye Sedef Keskin

6. Appendices

6.1. Appendix A Requirements Engineering Risks & Risk Management Plan